

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 2- Concept Mapping**

|                  |                                                                                                                                                                                                                                                                                                                           |               |                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | CSA0311                                                                                                                                                                                                                                                                                                                   | Course Title: | Data Structures                     |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                                                                                                                                                                                                                                | Assessment:   | Assessment Tool 2 – Concept Mapping |
| Weightage:       | 10% of CIA                                                                                                                                                                                                                                                                                                                | Total Marks:  | 100                                 |
| CO1 Statement:   | Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis. |               |                                     |
| Student Name:    | Mosikyan                                                                                                                                                                                                                                                                                                                  | Reg. No.:     | 192511353                           |
| Section / Batch: | 2025-2029                                                                                                                                                                                                                                                                                                                 | Date:         | 13/7/26                             |

**Code of Conduct**

I 192511353 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Mosikyan

**Instructions**

- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

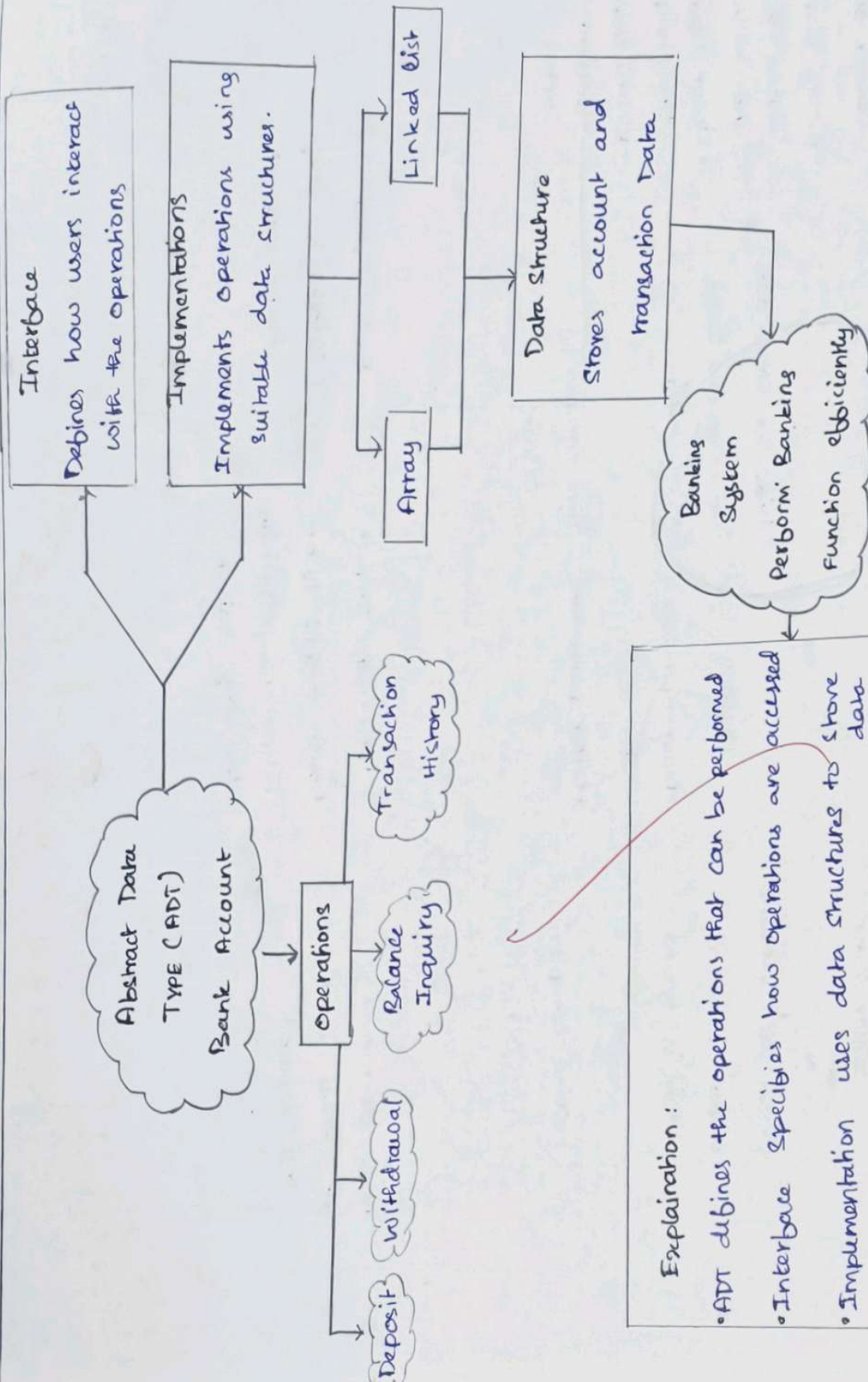
**Rubrics**

| Questions | Problem Understanding | Method Selection | Solution Process | Accuracy of Calculation | Interpretation of Results | Total Marks |
|-----------|-----------------------|------------------|------------------|-------------------------|---------------------------|-------------|
| Q1        |                       |                  |                  |                         |                           |             |
| Q2        |                       |                  |                  |                         |                           |             |



## ASSESSMENT - 01

A banking system performs operations such as deposits, withdrawals and transaction tracking.



## Explanation :

- ADT defines the operations that can be performed
- Interface specifies how operations are accessed
- Implementation uses data structures to store data
- This separation improves modularity, reusability and maintainability of system



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## Searching Algorithm

Explanation:

- Linear search checks element one by one.
- Binary search repeatedly divides the sorted data into halves.
- Linear search has  $O(n)$  time complexity.
- Binary search is more efficient for large sorted datasets.

